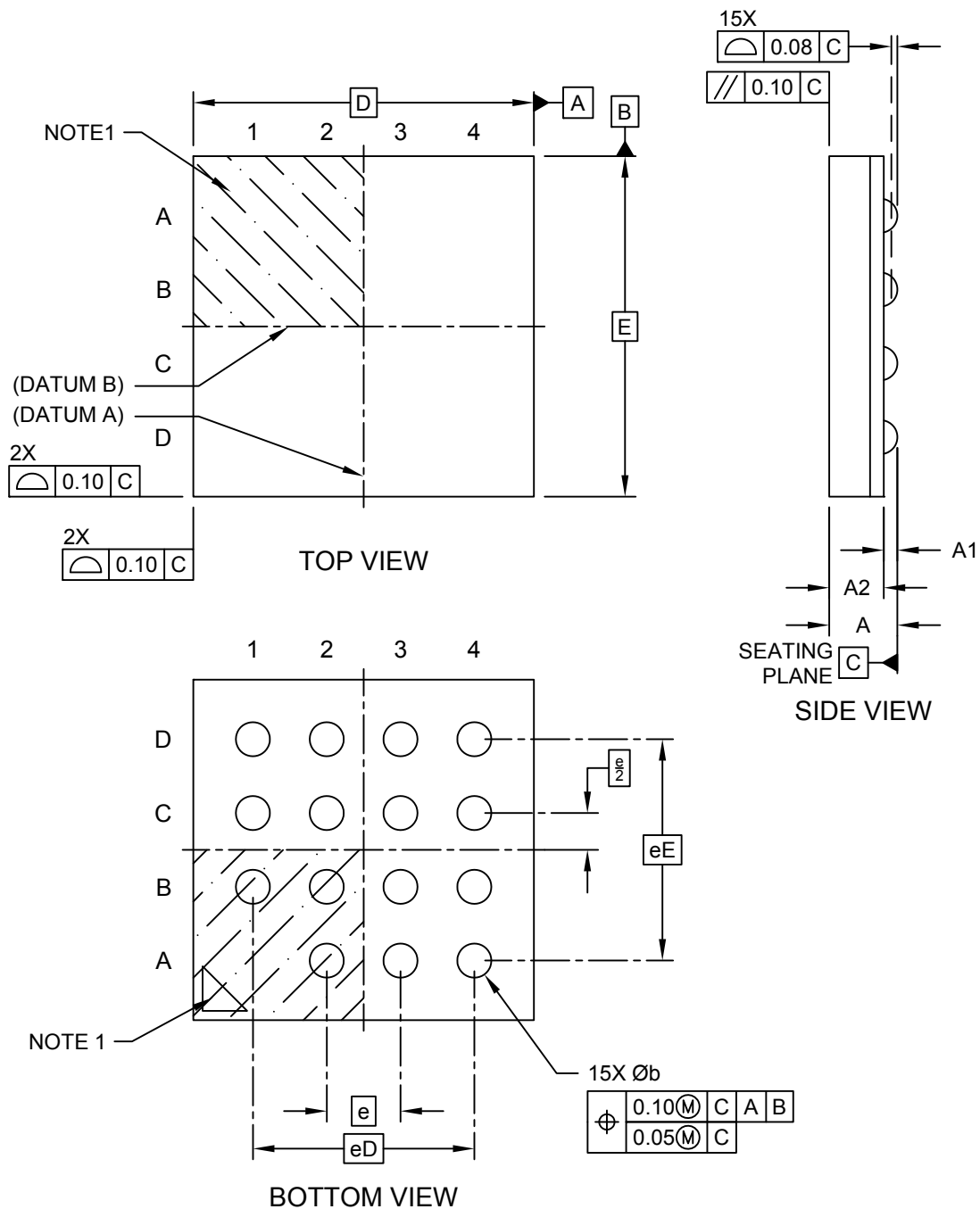


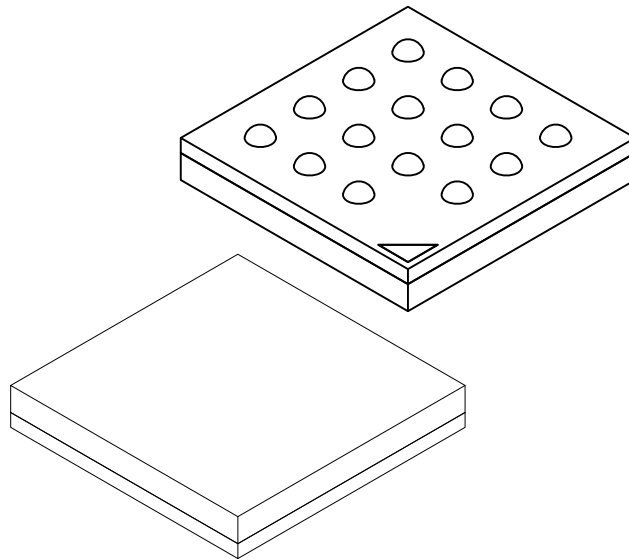
15-Ball Ultra Thin Fine Pitch Ball Grid Array (BNB) - 3x3x0.6 mm Body [UFBGA] Atmel Legacy Global Package Code CBC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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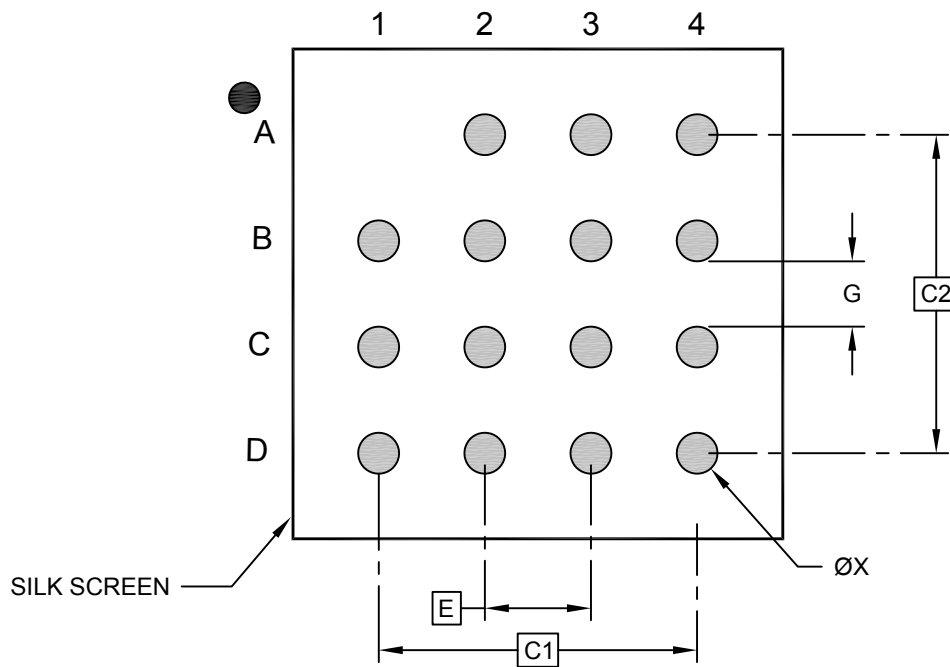
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	15		
Pitch	e	0.65 BSC		
Overall Terminal Spacing	eD	1.95 BSC		
Overall Terminal Spacing	eE	1.95 BSC		
Overall Height	A	–	–	0.60
Standoff	A1	0.12	–	–
Package Thickness	A2	0.38 REF		
Overall Length	D	3.00 BSC		
Overall Width	E	3.00 BSC		
Terminal Width	b	0.25	0.30	0.35

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1	1.95 BSC		
Contact Pad Spacing	C2	1.95 BSC		
Contact Pad Diameter (X15)	X			0.25
Contact Pad to Contact Pad	G	0.40		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23151 Rev A